

MRUDULA BALAGONI

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CAREER OBJECTIVES

Passionate and Detail-Oriented Computer Science graduate with a solid foundation in **software development, machine learning and cloud technologies**. Known for combining strong communication skills with technical expertise in **Python, SQL, Power BI** delivering real-world technical solutions. Seeking for an **Entry-level IT role** where I can leverage my skills in a company that values innovation and provides opportunities in professional growth and advancements.

EDUCATION

Master of Science (MS) in Computer Science | University of Alabama at Birmingham (UAB) Jan 2024 – May 2025
Relevant Courses: Machine Learning, Database System, Cloud Computing, Advanced Algorithms, Network Security, Cloud Security, Computer Security, Deep Learning, Foundations of Data Science.

Bachelor of Technology in Computer Science | Kakatiya Institute of Technology and Science (KITSW) Aug 2019 – May 2023
Relevant Courses: C, JAVA, Advanced Data Structures, Database Management System, Operating Systems, Python, Computer Networks, Software Engineering, Compiler Design, Cryptography, Data Mining, IOT, Machine Learning, Cyber Security
MDF Club Event Organizer who planned and managed club events and activities, handling details, leading teams, and creating a fun, inclusive environment for students.

EXPERIENCE

Supervisor in Recreation Center, UAB May 2024 – Jan 2025

- Advanced from Operations Attendant to Supervisor, showing leadership by guiding staff, managing daily tasks, and maintaining a safe, welcoming environment for patrons.
- Managed shift schedules, resolved issues, and maintained facility standards and Demonstrated leadership, teamwork, and strong communication under pressure.

Research Assistant at UAB – Machine Learning Jan 2025 – April 2025

- Applied Basic ML models to EHR data to predict patient outcomes. Cleaned, organized, and prepared real-world healthcare data for analysis using Python and SQL to make it suitable for machine learning pipelines.
- Worked closely with faculty and graduate researchers to review findings to interpret model outputs in a clinical context. Contributed to visualizing key results for research reports and academic presentations.

TECHNICAL SKILLS

Programming Languages: Python, JAVA, C, C++
Web and Database: HTML, CSS, JavaScript, MySQL,
Data Tools and Platforms: Power BI, Tableau, Microsoft Office, PowerPoint, Excel, Git

PROJECTS

Voice Based Email for Visually Impaired People Mar 2022 – June 2022

- Developed a Python-based application that enables visually impaired users to send, read, and manage emails entirely through voice commands, eliminating the need for keyboard or mouse interaction.
- Integrated speech-to-text and text-to-speech technologies provide a fully accessible email experience with support for sending messages and retrieving unseen emails via voice.

Phishing Website Detection Oct 2022 – Feb 2023

- Developed a phishing website detection system using supervised machine learning techniques, achieving 86.4% accuracy with the XGBoost model.
- Collected and preprocessed 10,000 URLs (phishing and legitimate), extracting 17 key features from URL structure and webpage content and evaluating multiple models including Decision Tree, SVM, and Neural Networks.

Credit Card Fraud Detection Jul 2024 – Dec 2024

- Developed a deep learning model using Multi-Layer Perceptron to detect fraudulent credit card transactions with 99.95% accuracy, addressing data imbalance with class weighting and SMOTE.
- Preprocessed anonymized PCA-transformed data and tuned hyperparameters to optimize model precision (82.35%) and recall (79.17%) and achieved effective fraud detection using TensorFlow/Keras.

Churn Prediction for Subscription Services Jan 2025 – April 2025

- Built machine learning models to predict customer churn using telecom service data. Preprocessed and analyzed behavioral, billing, and contract data to identify churn patterns and best model achieved an F1-score of 0.568 and ROC-AUC of 0.834.
- Improved churn prediction by addressing class imbalance and evaluating trade-offs between precision and recall for business impact.

CERTIFICATION

Oracle Academy – Database Design, Database Programming with SQL
Cisco Network Academy - Introduction of Cybersecurity, Cybersecurity Essentials
American Red Cross - First Aid/ CPR/AED